

ACCIDIOX — Video Script & Presentation Blueprint

Complete word-for-word demo script, technical breakdown, and pitch strategy designed to impress an IIT evaluation panel.

1 Strategic Alignment: What IIT Judges Look For

1. Physics & Math Validation

Gravity vector magnitude gating & 10-second sustained hold algorithm — eliminating false triggers from potholes & braking.

2. Full-Loop Working Execution

MPU-6050 Sensor → ESP32 Dual-Core Firmware → Real-Time BLE → Mobile PWA → Meta Cloud API → WhatsApp SOS.

3. Golden Hour & Safety UI

20-second safety window, "I Am Safe" false-positive cancellation, and 2-second periodic buzzer chime.

4. Real-World Commercialization

Direct OEM integration into EV/ICE CAN bus, 112 ERSS ambulance routing, and insurance telematics (UBI).

2 Video Timeline Overview (Target: 3:30 – 3:45 Mins)

Time	Segment	Visual Focus	Core Objective
0:00 – 0:30	Hook & Problem Reality	CCTV Crash Footage	Emotional hook: rider unable to call for help.
0:30 – 0:55	The Golden Hour Gap	Data & Fatality Charts	Statistical proof: 50% lives saved within 10 mins.
0:55 – 1:15	Introducing Accidiox	Hero Hardware Close-up	Unique tagline and product identity.
1:15 – 2:30	Full Live Working Demo	Hardware + Mobile App	10s hold, 2s buzzer, 20s window, WhatsApp SOS.
2:30 – 3:05	Engineering Architecture	System Architecture Diagram	ESP32 dual-core, BLE GATT, Meta Cloud API.
3:05 – 3:45	Future Scope & Impact	EV OEM & 112 ERSS Mockup	Mass manufacturing and public safety vision.

3 Word-for-Word Video Script & Visual Storyboard

SCENE 1: The Hook & Problem Reality

0:00 – 0:30

Visual: Quick dramatic cuts of real two-wheeler crashes. Abrupt cut to black with on-screen text: *"In a two-wheeler accident, the rider often cannot call for help."*

VOICEOVER (SERIOUS & AUTHORITATIVE TONE):

*"Every four minutes, a life is lost on Indian roads — and over 70% of all fatal road victims are two-wheeler riders. When a rider crashes in an isolated area, late at night, or falls unconscious, the biggest killer isn't just the impact — it is the catastrophic delay in emergency medical response. In trauma care, this critical window is known as the **Golden Hour**."*

SCENE 2: The Data & The Golden Hour Gap

0:30 – 0:55

Visual: Graphs showing MoRTH data: 1.7+ Lakh annual fatalities; 44.5% two-wheelers; 50% survival drop after 30 minutes.

VOICEOVER:

*"Official data from the Ministry of Road Transport reveals that nearly 50% of crash victims could have survived if emergency services received exact coordinates within the first ten minutes. While luxury four-wheelers come equipped with automated eCall crash systems, 250 million two-wheeler riders in India navigate traffic with **zero automated crash detection**."*

SCENE 3: Introducing Accidiox (The Winning Pitch Line)

0:55 – 1:15

Visual: Studio hero shot of the Accidiox hardware unit with OLED display and LEDs glowing on a clean workbench.

VOICEOVER (CONFIDENT & INSPIRING):

*"To solve this critical challenge, we built **Accidiox**."*

'Accidiox transforms any two-wheeler into an intelligent, connected life-saving vehicle — detecting crashes autonomously, eliminating false alarms through physics-driven filtering, and dispatching multi-channel emergency rescue within seconds.' "

SCENE 4: Full Live Working Demonstration

1:15 – 2:30

Demonstration 4A: Anti-False-Positive Filtering (0:15)

Visual: Vigorously shake the unit and simulate a bump. Show OLED/BLE remains normal. Tilt for 3 seconds then right it up — show no alarm.

VOICEOVER:

*"Traditional tilt sensors fail due to potholes, braking vibrations, and speed breakers. Accidiox eliminates false positives using **gravity-vector magnitude gating** and a **10-second continuous hold filter**. If the bike is righted within 10 seconds, the candidate resets completely — zero false alarms."*

Demonstration 4B: Sustained Crash & 2-Second Periodic Alert (0:25)

Visual: Keep the unit fallen past 85° for 10s. The hardware buzzer chirps every 2s, hazard LED pulses, and phone

triggers the 20s Emergency Window.

VOICEOVER:

"When a genuine fall is held past 10 seconds, the hardware confirms the crash: the onboard buzzer pulses every 2 seconds to alert bystanders without continuous noise fatigue, and broadcasts telemetry to our mobile app. The rider gets a 20-second safety window with a one-tap 'I Am Safe' override."

Demonstration 4C: Autonomous WhatsApp SOS & Database Sync (0:35)

Visual: 20s timer expires. Pop-up appears showing 'WhatsApp SOS Sent'. Switch to receiver phone showing incoming WhatsApp message with live Google Maps link and nearest hospital route.

VOICEOVER:

"If the timer expires, Accidiox autonomously dispatches an emergency SOS via the Meta WhatsApp Cloud API directly to all family contacts — delivering live GPS coordinates, incident timestamp, and the nearest hospital route. Simultaneously, the incident is securely logged to our cloud database and live Incident Map."

SCENE 5: Engineering Architecture & Tech Stack

2:30 – 3:05

Visual: Clean split-screen or animated architecture diagram highlighting the 3 layers.

VOICEOVER:

"Our engineering architecture is built across three production-grade tiers:

- 1. Hardware Layer:** ESP32 dual-core SoC paired with an MPU-6050 6-DOF IMU, running real-time Euler angle fusion and low-latency BLE GATT.
- 2. Frontend Layer:** A modern Progressive Web App (PWA) utilizing the Web Bluetooth API for live bi-directional telemetry streaming and Web Audio synthesis.
- 3. Cloud & Telematics Layer:** Hosted backend leveraging the Meta WhatsApp Graph API (v20.0), Geoapify Places API for automated hospital geocoding, and a MySQL relational database for audit-proof incident tracking."

SCENE 6: Future Scope & Real-World Scaling

3:05 – 3:45

Visual: Renders/mockups showing Accidiox integrated into electric scooters and centralized ERSS emergency dispatch dashboards.

VOICEOVER (STRONG CONCLUSION):

"Looking ahead, Accidiox is architected for mass commercialization:

- **OEM Integration:** Seamless embedded integration into EV and ICE two-wheeler CAN bus and OBD-II systems.
- **National 112 ERSS Dispatch:** Direct integration with India's Emergency Response Support System for automated ambulance routing.
- **Insurance Telematics:** Tamper-proof crash blackbox logs for Usage-Based Insurance (UBI) and accident reconstruction.

With Accidiox, we are ensuring that no rider on Indian roads is ever left alone in the Golden Hour. Thank you!"

4 Practical Filming & Recording Checklist

Production Area	Pro-Tip for Maximum Polish
Audio Recording	Use a dedicated lapel mic or wired headset mic. Ensure zero room echo. Add very soft, low-volume background tech music.
Camera Shots	Dual-angle setup: Wide angle showing presenter + Top-down close-up on the ESP32 board and buzzer LED.
Screen Recording	Record the smartphone screen with 60fps in dark mode. Show the 20s countdown and receiver phone side-by-side.
Picture-in-Picture	During the demo, show the physical hardware tilting on the left, and the mobile dashboard reacting live on the right.